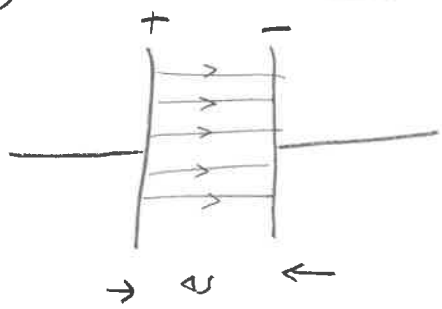
1. Chapter 25



$\Delta s = 5.40\ mm$

a) $\Delta V = 600$ Volts

$E_s = -\frac{\Delta V}{\Delta s}$ $\quad (\vec{E} = -\vec{\nabla} V)$

$|\vec{E}| = \frac{\Delta V}{\Delta s} = \frac{600\ V}{5.40\ mm} = 1.1 \times 10^5\ \frac{V}{m}$

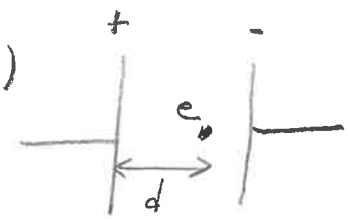
$= \boxed{1.1 \times 10^5\ \frac{N}{C}}$

b) $\vec{F} = q\vec{E}$, $\quad |\vec{F}| = eE$

$= 1.6 \times 10^{-19}\ C \left(1.1 \times 10^5\ \frac{N}{C}\right)$

$= \boxed{1.78 \times 10^{-14}\ N}$

c)



$W = F\,\Delta x$

$= 1.78 \times 10^{-14}\ N \cdot 2.48 \times 10^{-3}\ m$

$= \boxed{4.41 \times 10^{-17}\ J}$

$d = 2.92\ mm$

$\Delta s = 5.40\ mm$

$\Delta x = \Delta s - d = 2.48\ mm$

NOTE: $W = 4.41 \times 10^{-17}\ J \left(\frac{1\ eV}{1.602 \times 10^{-19}\ J}\right) = 276\ eV$

$\Delta V \left(\frac{2.48\ mm}{5.40\ mm}\right) = 276\ eV$ (same)

Recall: $1\ eV$ = amount of work required to move an electron thru a potential difference of 1 Volt.

(2.) a) Electron moving with speed $v = 2.75 \times 10^7$ m/s
$m_e = 9.11 \times 10^{-31}$ kg

$K = \frac{1}{2} m v^2$

$W_{net} = \Delta K$, $W_{net} = q \Delta V_e$

Thus, $\Delta V_e = \frac{W_{net}}{q} = \frac{-\frac{1}{2} m_e v^2}{-e} = \frac{\frac{1}{2}(9.11\times10^{-31})(2.75\times10^7)^2}{1.602\times10^{-19} C}$

$= \boxed{2.15 \times 10^3 \text{ V}}$

b) Proton moving at same speed v, but mass
Has same magnitude charge q
is $\approx 2000\times$ larger: $m_p = 1.67 \times 10^{-27}$ kg

Thus, $\Delta V_p \approx 2{,}000 \times \Delta V_e$

c) $\frac{\Delta V_p}{\Delta V_e} = \frac{\frac{1}{2} m_p \cancel{v^2}}{\cancel{q_p}} \frac{\cancel{q_e}}{\frac{1}{2} m_e \cancel{v^2}}$ $(q_p = q_e = e)$

$= \boxed{\frac{m_p}{m_e}}$

(3.) a) $r = 0.100$ cm from electron

$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ $(V(r=\infty) = 0)$

$q = -e = -1.602 \times 10^{-19}$ C

$\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{N \cdot m^2}$

$\rightarrow V(r = 0.100 \text{ cm}) = \boxed{-1.44 \times 10^{-6} \text{ V}}$

b) $r_1 = 0.100$ cm , $r_2 = 0.870$ cm

$V(r_1) = -1.44 \times 10^{-6}$ V

$V(r_2) = -1.66 \times 10^{-7}$ V

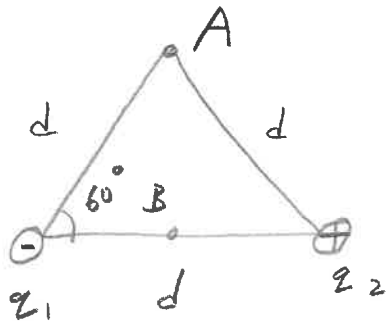
$\Delta V = V_2 - V_1 = \boxed{1.27 \times 10^{-6} \text{ V}}$

c) If electron replaced by a proton

a) sign change
b) sign change

} so both answer would change

4.



$d = 3.50$ cm

$q_1 = -19$ nC

$q_2 = 28.5$ nC

a) $$V(A) = \frac{k_e q_1}{d} + \frac{k_e q_2}{d}$$

$$= \frac{k_e}{d}(q_1 + q_2)$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{(3.5\times10^{-2}\,m)}(-19+28.5)\times10^{-9}\,C$$

$$= \boxed{2.44\times10^{3}\ \text{Volts}} = \boxed{2.44\ kV}$$

b) $$V(B) = \frac{k_e q_1}{d/2} + \frac{k_e q_2}{d/2}$$

$$= \frac{2k_e}{d}(q_1 + q_2)$$

$$= 2V(A)$$

$$= \boxed{4.88\times10^{3}\ \text{Volts}} = \boxed{4.88\ kV}$$

(5.) (4)

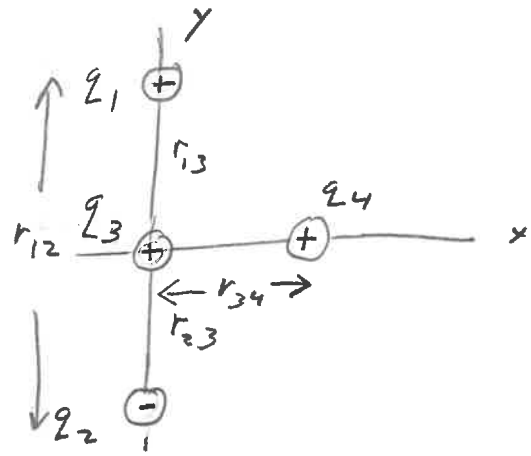
$q_1 = 90.0\,nC$, $q_2 = -90.0\,nC$,

$(x_1, y_1) = (0, 12.00\text{ cm}) = (0, y)$

$(x_2, y_2) = (0, -12.00\text{ cm}) = (0, -y)$

Let $Q = 90\,nC$
Then
$q_1 = Q$, $q_2 = -Q$
$q_3 = Q/2$

$q_3 = 45\,nC$ located at $(x_3, y_3) = (0, 0)$



a) $$U_3 = \frac{k_e q_1 q_2}{r_{12}} + \frac{k_e q_1 q_3}{r_{13}} + \frac{k_e q_2 q_3}{r_{23}}$$

$$= k_e\left(\frac{-Q^2}{2y} + \cancel{\left(\frac{Q^2/2}{y}\right)} + \cancel{\left(\frac{-Q^2/2}{y}\right)}\right)$$

$$= \frac{-k_e Q^2}{2y}$$

$$= \boxed{3.03 \times 10^{-4}\ J}$$

b) $m = 2.34 \times 10^{-13}\ kg$

$q_4 = 180\,nC = 2Q$

$r_{34} = 9\text{ cm} = \frac{3}{4}y$

$$U_4 = U_3 + \left(\frac{k_e q_1 q_4}{r_{14}} + \frac{k_e q_2 q_4}{r_{24}} + \frac{k_e q_3 q_4}{r_{34}}\right)$$

$$= \frac{-k_e Q^2}{2y} + \cancel{\frac{k_e(2Q^2)}{\sqrt{x^2+y^2}}} \cancel{\frac{-k_e(2Q^2)}{\sqrt{x^2+y^2}}} + \frac{k_e\left(\frac{Q}{2}\cdot 2Q\right)}{x}$$

$$= \frac{-k_e Q^2}{2y} + \frac{k_e Q^2}{x}$$

$$= -k_e Q^2\left(\frac{1}{2y} - \frac{1}{x}\right)$$

$$= \boxed{3.84 \times 10^{-4}\ J}$$

Initially,

$$E = U_4 \qquad (K = 0)$$

Finally,

$$E = U_3 + K$$

Thus, $U_4 = U_3 + K$

$$K = U_4 - U_3$$

$$= \frac{k_e Q^2}{x}$$

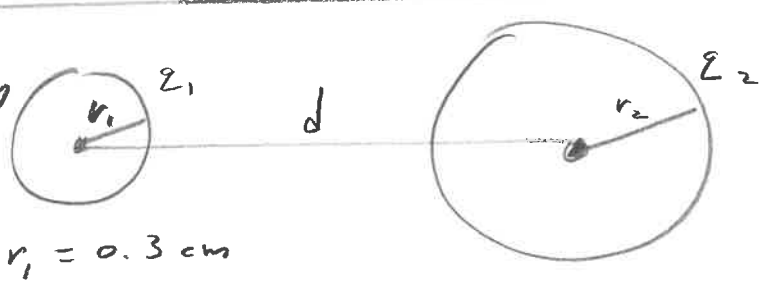
$$= 8.1 \times 10^{-5}\ J$$

Also, $K = \frac{1}{2} m v^2$

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(8.1 \times 10^{-5} J)}{2.34 \times 10^{-13}\ kg}}$$

$$= \boxed{2.6 \times 10^{4}\ \frac{m}{s}}$$

(6.) Two insulating spheres



$d = 1\ m$ (initial separation)

$r_1 = 0.3\ cm$
$m_1 = 0.4\ kg$
$q_1 = -2\ \mu C$

$r_2 = 0.5\ cm$
$m_2 = 0.7\ kg$
$q_2 = 5.5\ \mu C$

a) Initially: $E = U_i = \frac{k_e q_1 q_2}{d}$

Finally: $E = U_f + K_1 + K_2$

$$= \frac{k_e q_1 q_2}{r_1 + r_2} + \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Initially: $\vec{p}_{tot} = \vec{0}$

Finally: $\vec{p}_{tot} = m_1 v_1 \hat{x} + m_2 v_2 (-\hat{x})$

$= (m_1 v_1 - m_2 v_2)\hat{x}$

Thus, $m_1 v_1 = m_2 v_2 \rightarrow \boxed{v_2 = \left(\frac{m_1}{m_2}\right) v_1}$ (Cons. of energy)

$$\frac{k_e q_1 q_2}{d} = \frac{k_e q_1 q_2}{(r_1 + r_2)} + \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$ (Cons. of total momentum)

$$k_e q_1 q_2 \left(\frac{1}{d} - \frac{1}{r_1 + r_2}\right) = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 \left(\frac{m_1}{m_2}\right)^2 v_1^2$$

$$= \frac{1}{2} m_1 v_1^2 \left[1 + \left(\frac{m_1}{m_2}\right)\right]$$

$$= \frac{1}{2} m_1 v_1^2 \left(\frac{m_1 + m_2}{m_2}\right)$$

Thus, $$v_1^2 = \frac{2 m_2}{m_1 (m_1 + m_2)} k_e q_1 q_2 \left(\frac{r_1 + r_2 - d}{d(r_1 + r_2)}\right)$$

$$\Rightarrow \quad v_1 = \sqrt{\frac{-2 m_2/m_1}{(m_1 + m_2)} k_e q_1 q_2 \left(\frac{d - (r_1 + r_2)}{d(r_1 + r_2)}\right)}$$

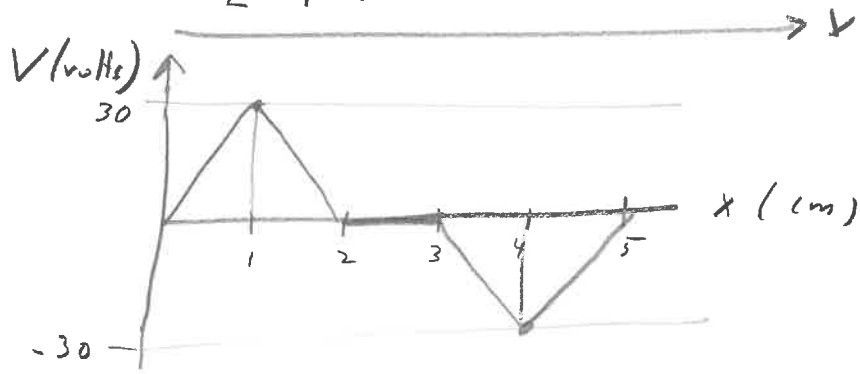
$$= \boxed{6.25 \text{ m/s}}$$

$$v_2 = \left(\frac{m_1}{m_2}\right) v_1 = \boxed{3.57 \text{ m/s}}$$

(b) Speeds would be the same if spheres were conductors instead of insulators.

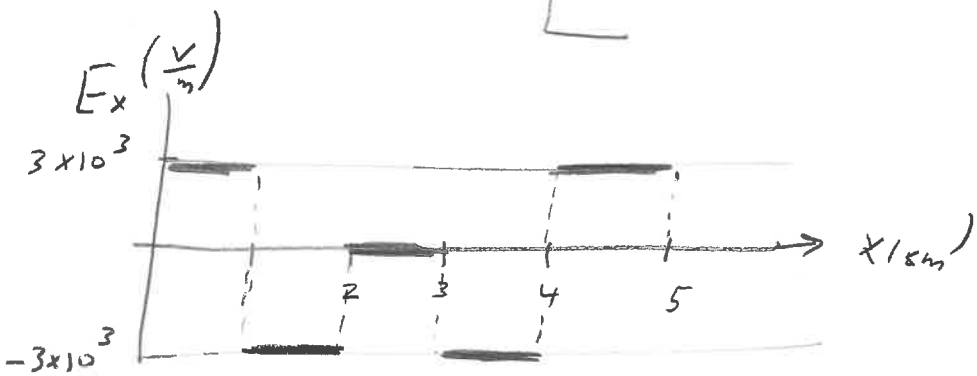
7.

$\vec{E}$ parallel to $\hat{x}$

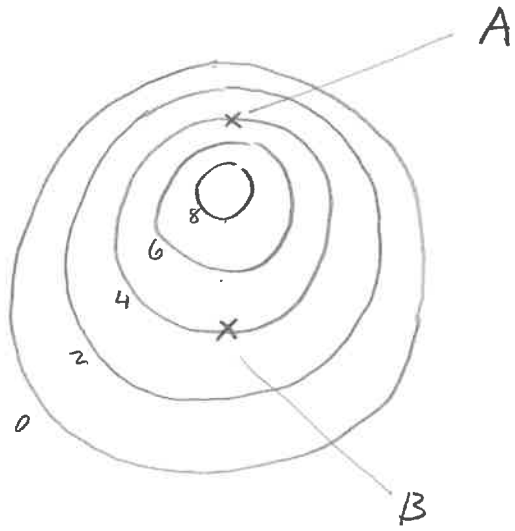


$$\vec{E} = -\vec{\nabla} V$$

$$E_x = -\frac{dV}{dx} = \begin{cases} \dfrac{30 \text{ volts}}{1 \text{ cm}} = 3\times10^3 \dfrac{V}{m} & (0 \le x \le 1\text{cm}) \\ -\dfrac{30 \text{ volts}}{1\text{cm}} = -3\times10^3 \dfrac{V}{m} & (1\text{cm} \le x \le 2\text{cm}) \\ 0 & (2\text{cm} \le x \le 3\text{ cm}) \\ -3\times10^3 \dfrac{V}{m} & (3\text{cm} \le x \le 4\text{ cm}) \\ +3\times10^3 \dfrac{V}{m} & (4\text{ cm} \le x \le 5\text{ cm}) \end{cases}$$



(8.)



$d = 1.50$ cm (separation between grid lines)

a) $E = -\frac{\Delta V}{\Delta s}$

$\Delta V = -2$ volts (for both A, B)

$\Delta s = \begin{cases} \frac{d}{2} & \text{(at A)} \\ d & \text{(at B)} \end{cases}$

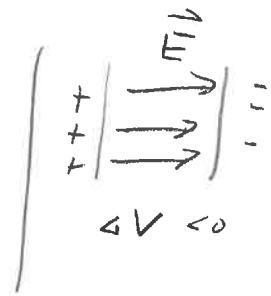
Thus, $|E_A| > |E_B|$ since Δs is smaller

b) $\vec{E}_B = -\vec{\nabla} V$ —— direction of gradient

$= -\hat{y} \left| \frac{\Delta V}{\Delta s} \right|$

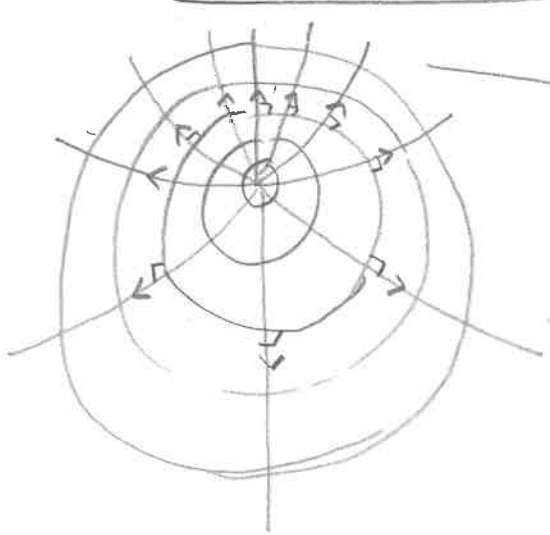
$= -\hat{y} \left| \frac{-2 \text{ volts}}{1.5 \text{ cm}} \right|$

$= \boxed{-133 \frac{\text{volts}}{\text{m}} \hat{y}}$



$\frac{2}{3/2} = \frac{4}{3} = 1.33$

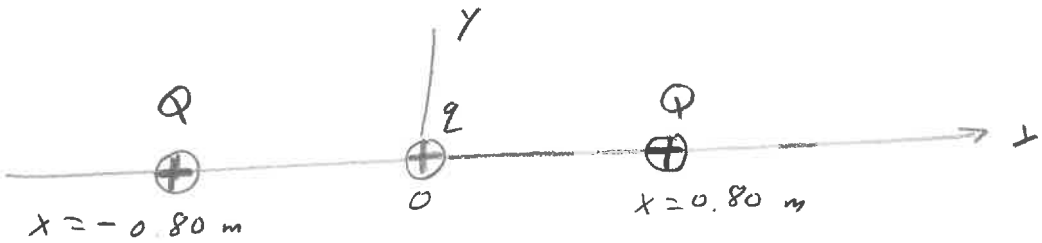
c)



density of field lines largest where equipotential lines are closest

Field lines $\perp$ to equipotential lines

(9.)



$Q = 2.10 \times 10^{-6}\ C$

$q = 1.20 \times 10^{-18}\ C$

a) $\vec{F}_{net,q} = \vec{0}$ since $\frac{k_e Q q}{x^2}(-\hat{x}) + \frac{k_e Q q}{x^2}(+\hat{x}) = 0$

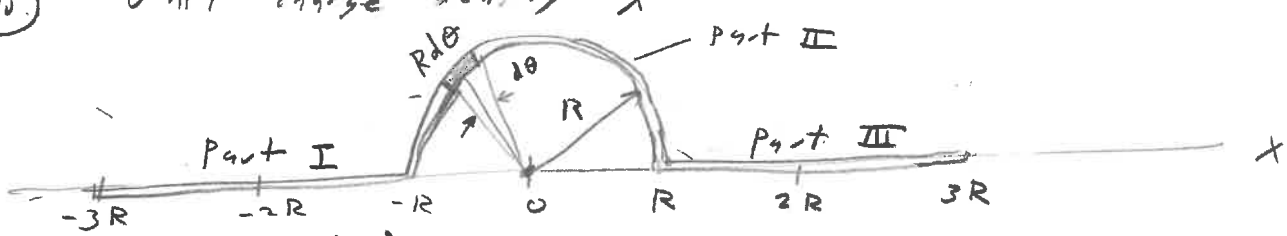
(same magnitude but opp direction)

b) $\vec{E} = \vec{0}$ since $\vec{E} = \frac{\vec{F}_{net,q}}{\vec{q}} = 0$

c) $V = \frac{k_e Q}{x} + \frac{k_e Q}{x} = \frac{2 k_e Q}{x} = 47 \times 10^3$ volts

$= \boxed{47\ kV}$

(10.) uniform charge density λ



Find $V(0)$

$V(0) = \int \frac{k_e\, dq}{r}$ $\qquad dq = \lambda\, ds$, $\begin{cases} ds = dx & \text{for Parts I, III} \\ ds = R\,d\theta & \text{for part II} \end{cases}$

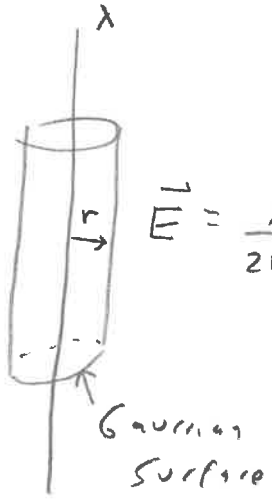
$$= \int_{-3R}^{-R} \frac{k_e \lambda\, dx}{|x|} + \int_{R}^{3R} \frac{k_e \lambda\, dx}{x} + \int_{0}^{\pi} \frac{k_e \lambda \cancel{R}\, d\theta}{\cancel{R}}$$

$$= k_e \lambda \left[2 \ln x \Big|_{R}^{3R} + \pi \right]$$

$$= k_e \lambda \left[2(\ln(3R) - \ln R) + \pi \right]$$

$$= k_e \lambda \left[2 \ln(3) + \pi \right]$$

(11.)



$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r}\hat{r}$$

(10)

$$V(r_2) - V(r_1) = -\int_{r_1}^{r_2} \vec{E}\cdot d\vec{s}$$

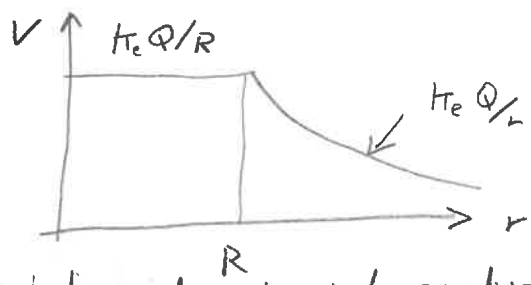
$d\vec{s} = dr\,\hat{r}$

$$= \frac{-\lambda}{2\pi\epsilon_0}\int_{r_1}^{r_2}\frac{dr}{r}$$

$$= -\frac{\lambda}{2\pi\epsilon_0}\ln r\Big|_{r_1}^{r_2}$$

$$= \boxed{-\frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{r_2}{r_1}\right)}$$

(12.)



$$E_r = -\frac{dV}{dr}$$

charged spherical conductor with total charge Q

a) Inside $(r \le R)$

$$E_r = -\frac{d}{dr}\left(\frac{k_e Q}{R}\right) = \boxed{0}$$

b) outside $(r > R$

$$E_r = -\frac{d}{dr}\left(\frac{k_e Q}{r}\right) = \boxed{\frac{k_e Q}{r^2}}$$